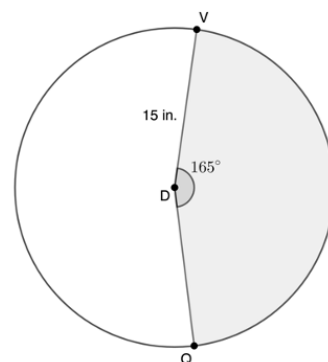


Geometry
Unit 10
Lesson 4 Practice

Name _____
Date _____
Period _____

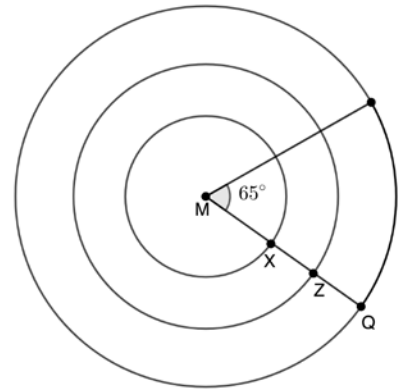
1. Circle C has a diameter of 18 inches (in.). Points T and S are on the circle and form the sector TCS . The central angle $\angle TCS$ in Circle C measures 120° . Determine the area of the sector. Express your answer in terms of π .
2. The area of a sector with a radius of 5 centimeters (cm) is 15.7 square cm. Calculate the approximate measure of the central angle of the sector in degrees.

3. Circle D has radius $\overline{DV}$ that measures 15 inches (in.) with central angle $\angle VDQ$, where $m\angle VDQ = 165^\circ$. Find the area of the shaded sector. Round your answer to the nearest hundredth.



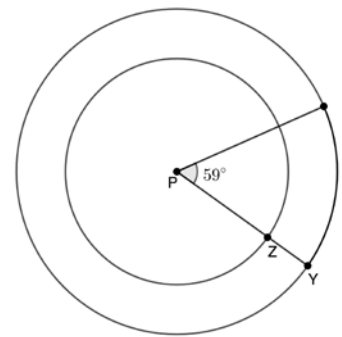
4. Circle K has a radius of 7 inches (in.). Points R and Y are on the circle and form the sector RKY . The central angle $\angle RKY$ in Circle K measures 145° . Determine the area of the sector. Express your answer in terms of π .
5. The area of a sector with a diameter of 22 feet (ft) is 27.44 square ft. Calculate the approximate measure of the central angle of the sector in degrees.

For questions 6-8, concentric circles with center M are shown with $m\angle M = 65^\circ$. The smallest Circle M has a radius $\overline{MX}$ that measures 10 yards (yd). The largest Circle M has radius $\overline{MQ}$ and the middle Circle M has radius $\overline{MZ}$. Each circle has a radius that is 10 yd longer than the immediate prior circle.



6. What is the arc length of the circle with radius $\overline{MX}$?
7. What is the arc length of the circle with radius $\overline{MZ}$?
8. What is the arc length of the circle with radius $\overline{MQ}$?

For questions 9-10, concentric circles with center P are shown with $m\angle P = 59^\circ$. The smallest Circle P has a radius $\overline{PZ}$ that measures 9 meters (m) and the largest Circle P has radius $\overline{PY}$. The radius of the largest Circle P has a radius 4 m larger than the smaller Circle P .



9. What is the arc length of the circle with radius $\overline{PZ}$?
10. What is the arc length of the circle with radius $\overline{PY}$?